

Final Project Manuscript

Mathematical Modeling of Chocolate Toxicity for Dogs

BS-3-S2-EC-PHYMAT / Systèmes physiologiques et modélisation mathématique

Team 6 – Third-Year Biotechnology and Bioinformatics, INSA Lyon

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Academic year 2025–2026

Role	Main responsibility	Student name
Physiology lead	Biological mechanism and physiological consistency	Marion Venrucci
Mathematics/modeling lead	Variables, equations, assumptions, analysis	Van Tran
Computation/simulation lead	Code, numerical simulations, figures	Capucine Schmidt
Critical assumptions lead	Assumptions, limitations, uncertainty	Olinda Mendes
Communication/report lead	Writing, structure, clarity	Doriane Rombaut
Coordinator	Organization, integration, deadlines	Coppélia Reynaert-Verbiese

Guidance.

Roles indicate responsibilities, not isolated tasks. Every student must understand the whole model and be able to explain the main biological and mathematical ideas.

Abstract

Write 150–200 words.

Your abstract should contain:

- the biological question;
- the modeling hypothesis;
- the type of mathematical model used;
- the main result;
- one important limitation.

Do not include technical details, long equations, or references in the abstract.

1 Biological and Modeling Question

1.1 Biological context

Chez le chien, c'est la théobromine qui est responsable de la toxicité du chocolat. En effet, une fois dans le système gastrique, elle est absorbée dans le sang et diffusée dans l'organisme, impactant ainsi des systèmes essentiels tels que le système nerveux ou cardiovasculaire.

La théobromine ayant un impact sur l'entièreté de l'organisme, nous avons choisi de prendre comme système le chien dans son ensemble. Ce système abrite ainsi l'évolution de la concentration de la théobromine dans le sang et son impact sur l'organisme. En effet, en augmentant, la quantité de théobromine entraîne une réaction du corps dont découle l'apparition de certains symptômes. C'est le lien entre cette concentration et ces signes naissants que nous cherchons à comprendre. Cette réaction peut être observée sur plusieurs heures.

1.2 Modeling question

State the precise modeling question.

Comment est-ce que la théobromine évolue au cours du temps et agit sur l'évolution des symptômes chez le chien ? A partir de quelle seuil la dose ingérée est-elle mortelle ?

1.3 Reference regime and perturbation

Pour répondre à cette question, nous prendrons comme régime de référence un modèle sain, c'est-à-dire un chien n'ayant pas ingéré de chocolat. La perturbation étudiée est l'entrée dans le système de chocolat, et donc de théobromine, à un instant t_0 .

2 System Boundary, Variables, Parameters, and Assumptions

Guidance.

This section defines the model before equations are written. A model with unclear variables or hidden assumptions is not scientifically controlled.

2.1 System boundary

State what is inside and outside the model.

- Dans le modèle :
 - Voies circulatoires et sang
 - Système digestif
 - Autres voies métaboliques indirectement impliquée : système nerveux, respiratoire...

- Organes
- Récepteurs, protéines, enzymes
- Hormones et neurotransmetteurs
- A l'extérieur du modèle :
 - Potentiels problèmes de santé antérieurs du chien
 - Température pouvant impacter l'activité du métabolisme
 - Le reste est-ce que c'est vraiment en dehors ?

2.2 State variables

Symbol	Meaning	Unit
$Q(t)$	Masse de théobromine présente dans le système digestif	mg
$D(t)$	Concentration de théobromine absorbée dans le sang par la masse corporelle	mg/kg
$S(t)$	Intensité des symptômes	%

2.3 Parameters

Symbol	Meaning	Unit
M_{choc}	<i>Masse de chocolat ingérée</i>	g
C_{type}	<i>Concentration en théobromine selon le type de chocolat</i>	mg/g
W	<i>Masse du chien</i>	kg
f	<i>Fraction de théobromine non évacuée directement par l'organisme</i>	%
k_{abs}	<i>Taux d'absorption de théobromine dans le sang</i>	h^{-1}
k_e	<i>Taux d'élimination de la théobromine</i>	h^{-1}
ρ	<i>Taux d'activation des symptômes</i>	h^{-1}
K_D	<i>Seuil d'activation des symptômes</i>	mg/kg
n	<i>Coefficient de Hill</i>	<i>Sans unité</i>
μ_S	<i>Taux de récupération des symptômes</i>	h^{-1}

2.4 Inputs and observables

Inputs. Nos variables d'entrée sont toutes celles liées à l'ingestion de chocolat. On compte ainsi :

- La masse de chocolat ingérée : M_{choc}
- La concentration en théobromine pour le type de chocolat ingéré : C_{type}
- La fraction de théobromine non évacuée directement par l'organisme : f

Inputs are imposed from outside the model, for example:

$$u(t), \quad D_{\text{ext}}(t), \quad S(t).$$

Describe your input protocol.

Observables. Observables are quantities computed from the model and interpreted biologically.

Examples:

$$X_{\text{max}}, \quad T_{\text{rec}}, \quad \int_0^T X(t) dt, \quad X(T).$$

List the observables used in your project.

2.5 Assumptions

Assumption	Reason	Possible limitation
<i>Assumption 1</i>	<i>Why it is made</i>	<i>Limit or bias</i>
<i>Assumption 2</i>	<i>Why it is made</i>	<i>Limit or bias</i>
<i>Assumption 3</i>	<i>Why it is made</i>	<i>Limit or bias</i>
<i>Assumption 4</i>	<i>Why it is made</i>	<i>Limit or bias</i>
<i>Assumption 5</i>	<i>Why it is made</i>	<i>Limit or bias</i>

3 Model Construction

Guidance.

Each equation must be readable as a biological sentence. Do not write a term unless you can explain what it represents.

3.1 Conceptual diagram

Insert your conceptual diagram.

Insert conceptual diagram here.

Figure 1: Conceptual diagram of the modeled system. Arrows indicate activation, production, transfer, degradation, inhibition, or feedback.

3.2 Model equations

Write the final model.

For example:

$$\frac{dX}{dt} = \text{production} - \text{clearance} + \text{activation} - \text{inhibition}.$$

Then give the actual equations:

$$\frac{dX}{dt} = F(X, Y, Z, t; \theta),$$

$$\frac{dY}{dt} = G(X, Y, Z, t; \theta),$$

$$\frac{dZ}{dt} = H(X, Y, Z, t; \theta).$$

Remove unused equations if your model has only one or two variables.

3.3 Meaning of each term

Explain each term.

For example:

$$-kX$$

represents linear clearance of X , with clearance rate $k > 0$.

$$\alpha \frac{Y^n}{K^n + Y^n}$$

represents saturating activation by Y , with maximal activation rate α , half-activation scale K , and Hill coefficient n .

3.4 Units and dimensional consistency

Check that every term in each equation has the same unit as the corresponding left-hand side.

For example, if X is a concentration, then

$$\left[\frac{dX}{dt} \right] = \frac{\text{concentration}}{\text{time}}.$$

Therefore every term in the equation for X must have units of concentration per time.

3.5 Biological domain

State the domain of the variables.

Examples:

$$X(t) \geq 0,$$

$$0 \leq B(t) \leq 1,$$

$$0 \leq V(t) \leq 1.$$

Explain whether the model preserves this domain.

4 Mathematical Analysis

Guidance.

*This is the mathematical core of the report. It should contain more than simulation.
At minimum, identify equilibria and discuss stability.*

4.1 Equilibria

Set the derivatives equal to zero.

For a one-variable model:

$$\frac{dX}{dt} = f(X),$$

equilibria satisfy

$$f(X^*) = 0.$$

For a two-variable model:

$$\frac{dX}{dt} = f(X, Y), \quad \frac{dY}{dt} = g(X, Y),$$

equilibria satisfy

$$f(X^*, Y^*) = 0, \quad g(X^*, Y^*) = 0.$$

List the equilibria and interpret them biologically.

4.2 Stability

For a one-dimensional model, compute

$$f'(X^*).$$

If

$$f'(X^*) < 0,$$

the equilibrium is locally stable.

If

$$f'(X^*) > 0,$$

the equilibrium is unstable.

For a two-dimensional model, compute the Jacobian:

$$J(X, Y) = \begin{pmatrix} \frac{\partial f}{\partial X} & \frac{\partial f}{\partial Y} \\ \frac{\partial g}{\partial X} & \frac{\partial g}{\partial Y} \end{pmatrix}.$$

Evaluate J at equilibrium and discuss trace, determinant, or eigenvalues.

4.3 Time scales

Identify the main time scales.

Examples:

$$\tau = \frac{1}{k},$$

or

$$t_{1/2} = \frac{\ln 2}{k}.$$

Interpret them biologically.

4.4 Thresholds, feedback, or bifurcations

Use this subsection only if relevant.

State clearly whether the project involves:

- an operational threshold;
- a dynamical threshold;
- positive feedback;
- negative feedback;
- bistability;
- hysteresis;
- bifurcation;
- irreversible loss.

Limitation.

Do not claim bistability, hysteresis, Hopf bifurcation, or point of no return unless your equations actually support that interpretation.

5 Simulation Protocol

Guidance.

A simulation is an experiment on the model. It must be reproducible.

5.1 Parameter values

Give the parameter set used for baseline simulations.

Parameter	Value	Unit	Justification
k			
α			
K			
n			

5.2 Initial conditions

State the initial conditions:

$$X(0) = X_0, \quad Y(0) = Y_0, \quad Z(0) = Z_0.$$

Explain their biological meaning.

5.3 Input protocol

Describe any external input.

Example:

$$u(t) = \begin{cases} u_0, & 0 \leq t \leq T_{\text{stim}}, \\ 0, & t > T_{\text{stim}}. \end{cases}$$

5.4 Numerical method

State the software and solver used.

Example:

The ODE system was integrated in Python using `solve_ivp` from `scipy.integrate`, with relative tolerance ... and absolute tolerance

or

The ODE system was integrated in MATLAB using `ode45`.

5.5 Simulation observables

Define the quantities extracted from simulations.

Examples:

$$X_{\max} = \max_{0 \leq t \leq T} X(t),$$

$$\text{AUC} = \int_0^T X(t) dt,$$

$$T_{\text{rec}} = \inf \{t \geq 0 : |X(s) - X^*| < \varepsilon \text{ for all } s \geq t\}.$$

6 Results

Guidance.

Each figure must answer a question. Do not include plots only because they look nice.

6.1 Baseline dynamics

Present the baseline simulation.

Insert baseline trajectory figure here.

Figure 2: Baseline dynamics of the model. The figure should show the main state variables and the relevant time scale.

Interpret the figure.

Questions:

- Does the system approach equilibrium?
- Is the response transient or persistent?
- Is there overshoot?
- What is the recovery time?
- Does the result match the stability analysis?

6.2 Perturbation or treatment scenario

Present the main perturbation scenario.

Insert perturbation or treatment figure here.

Figure 3: Effect of the selected perturbation on model dynamics.

Interpret the result.

6.3 Scenario comparison

Compare at least two scenarios.

Examples:

- low versus high exposure;
- weak versus strong feedback;

- fast versus slow clearance;
- early versus late treatment;
- different initial conditions.

7 Sensitivity and Robustness Analysis

Guidance.

This section shows whether your conclusion depends on one arbitrary parameter choice or remains true across a reasonable range.

7.1 One-parameter sweep

Choose one important parameter λ and vary it.

Define an observable $Q(\lambda)$.

Examples:

$$k \mapsto T_{\text{rec}}(k),$$

$$D \mapsto X_{\text{max}}(D),$$

$$\alpha \mapsto X(T; \alpha).$$

Insert parameter sweep figure here.

Figure 4: Effect of parameter variation on the selected observable.

7.2 Robustness of the main conclusion

State whether the main conclusion is robust.

Examples:

The conclusion remains valid when k varies over the interval $[k_{\text{min}}, k_{\text{max}}]$.

or

The conclusion depends strongly on the feedback strength α , so the result should be interpreted cautiously.

8 Discussion and Model Criticism

Guidance.

This section is not a place to repeat the results. It is where you explain what the model suggests and what it cannot prove.

8.1 Main biological interpretation

Explain the biological meaning of the results.

Main claim.

State your main claim in one or two sentences.

Example: The model suggests that recovery time is controlled primarily by the clearance rate, while peak response is controlled primarily by the input amplitude.

8.2 What the model supports

List the conclusions that are directly supported by your analysis and simulations.

- [Supported conclusion 1]
- [Supported conclusion 2]
- [Supported conclusion 3]

8.3 What the model does not prove

List conclusions that would be overclaims.

- [Overclaim 1]
- [Overclaim 2]

8.4 Limitations

Discuss limitations.

Possible limitations:

- simplified physiology;
- missing spatial structure;
- no delay;
- constant parameters;
- uncertain parameter values;

- no individual variability;
- no direct data fitting;
- phenomenological variables.

8.5 Possible improvements

State how the model could be improved.

Examples:

- adding a second compartment;
- adding a delay;
- using experimental data to estimate parameters;
- adding individual variability;
- testing an alternative feedback structure.

9 Conclusion

Write a short conclusion.

It should answer:

- What was the biological question?
- What was the model?
- What was the main mathematical result?
- What is the main biological interpretation?
- What is the main limitation?

A good conclusion is precise and modest.

Do not introduce new results here.

10 AI and Software Use Statement

10.1 AI use

State how AI tools were used.

Examples:

- brainstorming biological mechanisms;
- identifying hidden assumptions;
- checking clarity of explanations;

- debugging code;
- improving English formulation.

State clearly what was verified by the group.

Date	Task	Prompt or use	Group decision

10.2 Software use

State the software used.

Examples:

- Python;
- MATLAB;
- Jupyter Notebook;
- NumPy/SciPy;
- Matplotlib;
- Git or shared folder.

Acknowledgments

Optional.

Mention help received from instructors, classmates, or external sources if relevant.

References

- [1] S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. CRC Press.
- [2] L. Edelstein-Keshet, *Mathematical Models in Biology*. SIAM Classics in Applied Mathematics.
- [3] J. D. Murray, *Mathematical Biology I: An Introduction*. Springer.
- [4] B. P. Ingalls, *Mathematical Modeling in Systems Biology: An Introduction*. MIT Press.
- [5] U. Alon, *An Introduction to Systems Biology: Design Principles of Biological Circuits*. Chapman and Hall/CRC.
- [6] A. Saltelli et al., *Global Sensitivity Analysis: The Primer*. Wiley.

A Appendix A: Additional Calculations

Include long algebraic derivations, equilibrium calculations, nondimensionalized forms, or stability calculations that would interrupt the main text.

B Appendix B: Code Availability

State where the code is available.

Example:

The Python scripts and notebooks used for the simulations are included in the Moodle submission as supplementary files.

C Appendix C: Additional Figures

Include additional simulations or parameter sweeps if needed.